

Repository Analysis Report

for [Air-quality-AutoMLs](#)



- Repository Name: [Air-quality-Auto...](#)
- Owner: [ITMO-NSS-team](#)
- Created at: 01.05.2025 14:33

Statistics	Values
Stars Count	2
Forks Count	1
Issues Count	0

Metric	Values
README Presence	✓
License Presence	✓
Documentation Presence	X
Examples Presence	X
Requirements Presence	X
Tests Presence	X
Description Presence	✓

Repository Structure:

- Compliance: The repository adheres to a standard structure for data science projects, with clear separation of code, data, and results. The use of `.github` directory suggests an intention towards community contribution.
- Missing files: `requirements.txt`, `setup.py`, `tests/`
- Organization: The project is reasonably well-organized, with dedicated directories for baselines, data processing, visualization, and results. However, the lack of a formal requirements file or setup script hinders reproducibility.

README Analysis:

- Quality: Good. The README provides a clear overview of the project's goals, algorithms used, installation instructions, and citation information.
- Project description: Yes
- Installation: Yes
- Usage examples: No
- Contribution guidelines: Partial
- License specified: Yes
- Badges present: Yes

Documentation:

- Tests present: No
- Documentation quality: Limited. While the README provides some documentation, there is no dedicated documentation folder or more extensive guides.
- Outdated content: No

Key Shortcomings:

- Lack of a requirements file for reproducibility.
- Absence of unit tests to ensure code quality and prevent regressions.
- No clear examples on how to use the provided scripts or models.

Recommendations:

- Create a `requirements.txt` file listing all project dependencies.
- Implement a suite of unit tests using a testing framework like `pytest`.
- Add example notebooks demonstrating how to run the baseline models, AutoML pipelines and visualize results.
- Consider adding more detailed documentation on data preprocessing steps.
- Explore continuous integration/continuous deployment (CI/CD) practices.